

Stage 2 Hard-Negative Dataset Requirements

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Purpose

This dataset is for validating the **Stage 2 VLM wheel-mask filter**, not for training the first candidate segmentation model.

The Stage 2 pipeline is:

```
Roboflow / YOLO segmentation
-> wheel candidate masks with IDs
-> VLM sees original image + labeled overlay + candidate metadata
-> VLM returns keep/reject candidate IDs
-> backend combines selected masks
```

The dataset must answer one question:

Can the pipeline keep real physical wheels and reject reflections, shadows, background wheels, and other wheel-like artifacts?

Dataset Name

Recommended working name:

dream-wheels-stage2-hard-negatives

This should be separate from:

- Stage 1 training data;
- Stage 1.5 candidate sanity data;
- generated product outputs such as RESULTADOS/.

Minimum Size

For the next MVP validation pass:

30-50 images minimum

Recommended first useful target:

50 images

Pre-production target:

100-150 images

Required Image Buckets

For a 50-image MVP benchmark:

Bucket	Count	Why
Wet road / water reflections	10	Primary known failure mode.
Glossy floor / showroom reflections	8	Similar reflection geometry, different environment.
Night / low-light / neon glare	8	Hard visual conditions and bright highlights.
Shadows shaped like wheels	5	Prevent shadow masks from leaking into final output.
Background vehicles / background wheels	7	Avoid selecting wheels that are not on the user car.
Cropped / partial / occluded wheels	6	Common user-photo failure mode.

Bucket	Count	Why
Clean ordinary car photos	6	Baseline control group.

It is fine if one image belongs to multiple buckets, but the manifest should include all applicable tags.

Image Requirements

Each image should:

- contain a car or car-like target object;
- be representative of user-submitted Telegram photos where possible;
- preserve the original aspect ratio and resolution where possible;
- avoid heavy manual edits unless explicitly tagged as synthetic/generated;
- have clear licensing or consent status;
- be stored outside the Git repository unless it is a tiny curated example.

Preferred sources:

1. Real test-user/team photos with permission.
2. Parking-lot, street, showroom, garage, and wet-road photos collected for this benchmark.
3. Video frames from permitted videos.
4. Public internet images only when license/usage rights are clear.

Avoid mixing in generated AI outputs unless the image is explicitly tagged as synthetic.

Required Labels

This benchmark does not require full pixel-perfect annotation for every object at first. It does require reviewable labels for candidate decisions.

For each Roboflow candidate mask, label one of:

physical_wheel
reflection
shadow
background_wheel
body_or_tire_artifact
other_artifact
uncertain

At the image level, label:

has_reflection
has_shadow
has_background_vehicle
has_cropped_wheel
has_occlusion
lighting_condition
view_angle
expected_visible_wheel_count

Manifest Schema

Store the manifest in Git, but keep the binary image files in Roboflow, local storage, cloud storage, or Git LFS if needed.

Recommended CSV columns:

image_id,
source_uri,
source_type,
license_or_consent,
split,

```
tags,
lighting_condition,
view_angle,
expected_visible_wheel_count,
has_reflection,
has_shadow,
has_background_vehicle,
has_cropped_wheel,
has_occlusion,
notes
```

Candidate-level review output should use a second file:

```
image_id,
candidate_id,
candidate_class,
candidate_bbox_xyxy,
candidate_area_ratio,
model_keep,
model_confidence,
human_label,
review_notes
```

Evaluation Metrics

Primary Stage 2 metrics:

Metric	MVP Target
Physical wheel candidate keep accuracy	$\geq 90\%$
Reflection candidate rejection accuracy	$\geq 85\%$
Background wheel rejection accuracy	$\geq 85\%$
Image-level final mask acceptable rate	$\geq 85\%$
Critical failure rate	$\leq 5\%$

Critical failures:

- no physical wheel kept when a visible wheel candidate exists;
- reflection/shadow dominates the final mask;
- background vehicle wheel selected instead of the user car wheel;
- VLM returns invalid JSON or unusable candidate IDs.

Candidate Generator Requirements

Before evaluating VLM quality, verify candidate recall:

Roboflow should propose at least one candidate for each visible physical wheel in $\geq 90\%$ of benchmark images.

Important implementation rule:

Do not use `top_n=2` for Stage 2 filtering.

Pass all reasonable wheel candidates after class and minimum-area filtering.

The `cover.jpg` experiment showed that `top_n=2` can drop a small real rear wheel before the VLM sees it.

Model Testing Plan

Default model:

`qwen/qwen3-v1-30b-a3b-instruct`

Quality fallback:

qwen/qwen3-vl-235b-a22b-instruct

Evaluation flow:

Run Roboflow candidates

- > generate labeled overlay
- > run Qwen3-VL-30B
- > save JSON decision
- > manually review candidate labels
- > retry failure/low-confidence cases with Qwen3-VL-235B

Escalate to the larger model when:

- confidence is below threshold;
- no candidates are kept;
- all candidates are rejected despite expected visible wheels;
- reflection-like geometry is present;
- output JSON is invalid or incomplete.

What Counts As Done

Stage 2 MVP validation can be considered passed when:

1. The hard-negative dataset has at least 50 reviewed images.
2. Candidate-level labels exist for all Roboflow wheel candidates.
3. Qwen3-VL-30B meets the MVP metric targets.
4. Qwen3-VL-235B fallback improves most low-confidence or failed cases.
5. Remaining failures are documented with clear categories and next actions.

Current Status

Current status as of 2026-05-13:

- Stage 2 architecture is validated on the initial `cover.jpg` reflection case.
- Qwen3-VL-30B matched Qwen3-VL-235B on that case.
- Stage 1.5 candidate sanity dataset was created from `wheeltirebody/wheels-tires-body v2`.
- Stage 1.5 benchmark result: 125 images, 43 no-wheel negatives, 0 false positives, 3 false negatives.
- A dedicated Stage 2 hard-negative dataset has not been collected yet.

Conclusion:

The branch is ready to publish as a Stage 2 research/prototype milestone.

Full Stage 2 MVP validation requires the hard-negative dataset described here.